

Second-Order Elliptic PDE

Day 1

These notes develop the main calculations in detail, including clarifications of potentially confusing points.

1 From inverse-square fields to Poisson's equation

For a point mass m at $y \in \mathbb{R}^3$, the gravitational field at x is, up to the gravitational constant,

$$F_y(x) = m \frac{y - x}{|y - x|^3}.$$

Let $S_r(y)$ be a sphere of radius r centered at y . Its outward unit normal points away from y , whereas the gravitational field points toward y . Therefore

$$F_y \cdot \nu = -\frac{m}{r^2}$$

on the sphere. The total outward flux is

$$\int_{S_r(y)} F_y \cdot \nu \, dS = -\frac{m}{r^2} 4\pi r^2 = -4\pi m.$$

This is Gauss's law for a point mass. For a continuous mass density ρ , superposition gives

$$\mathbf{g}(x) = \int_{\Omega} \rho(y) \frac{y - x}{|y - x|^3} \, dy.$$

Gauss's law now states that the outward flux through the boundary of any smooth region D equals -4π times the enclosed mass:

$$\int_{\partial D} \mathbf{g} \cdot \nu \, dS = -4\pi \int_D \rho \, dx.$$

Apply the divergence theorem to the left side:

$$\int_D \operatorname{div} \mathbf{g} \, dx = -4\pi \int_D \rho \, dx.$$

Since this holds for every such region D ,

$$\operatorname{div} \mathbf{g} = -4\pi\rho.$$

The field is curl-free away from its sources and has a scalar potential u satisfying

$$\mathbf{g} = \nabla u.$$

One choice of potential is

$$u(x) = \int_{\Omega} \frac{\rho(y)}{|x-y|} dy$$

Combining $\mathbf{g} = \nabla u$ with Gauss's law gives Poisson's equation directly:

$$\Delta u = -4\pi\rho.$$

Equivalently, the point-source calculation gives the distributional identity

$$\Delta_x \frac{1}{|x-y|} = -4\pi\delta_y(x).$$

Where $\rho = 0$, the potential is harmonic:

$$\Delta u = 0.$$

[Definition] The Dirichlet problem on a bounded domain Ω is

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases}$$

The sign $-\Delta$ is common in analysis because it is nonnegative under homogeneous Dirichlet boundary conditions:

$$\int_{\Omega} u(-\Delta u) dx = \int_{\Omega} |\nabla u|^2 dx.$$

2 Radial harmonic functions

[Lemma] Suppose $u(x) = U(r)$ with $r = |x|$ in $\mathbb{R}^n$. Then

$$\Delta u = U''(r) + \frac{n-1}{r} U'(r).$$

One geometric derivation uses the shell between radii r and $r + dr$. Conservation of flux for a harmonic function gives

$$r^{n-1} U'(r) = \text{constant},$$

equivalently

$$\left(r^{n-1}U'(r)\right)' = 0.$$

Therefore

$$U(r) = \begin{cases} A \log r + B, & n = 2, \\ Ar^{2-n} + B, & n \geq 3. \end{cases}$$

If $n \geq 3$ and $U(r) \rightarrow 0$ as $r \rightarrow \infty$, then $B = 0$. The singular term r^{2-n} is harmonic away from the origin and becomes a fundamental solution after normalization.

3 Fundamental solution of the Laplacian

[Definition] Let $\omega_n = |S^{n-1}|$ be the area of the unit sphere. For the convention

$$-\Delta \Gamma = \delta_0,$$

we first interpret the equation as Gauss's law. Define the flux field

$$F = -\nabla \Gamma.$$

Then

$$\operatorname{div} F = -\Delta \Gamma = \delta_0.$$

Thus the origin is a unit point source. Let B_r be a ball centered at the origin. Gauss's law says that the total outward flux through every sphere enclosing the origin is one:

$$\int_{\partial B_r} F \cdot \nu \, dS = 1.$$

This is also obtained directly from the divergence theorem:

$$\int_{\partial B_r} F \cdot \nu \, dS = \int_{B_r} \operatorname{div} F \, dx = \int_{B_r} \delta_0 \, dx = 1.$$

Write $\Gamma(x) = U(r)$ with $r = |x|$. On ∂B_r the outward normal is radial, so

$$F \cdot \nu = -\nabla \Gamma \cdot \nu = -U'(r).$$

Since the area of ∂B_r is $\omega_n r^{n-1}$, Gauss's law becomes

$$-U'(r)\omega_n r^{n-1} = 1.$$

Hence

$$U'(r) = -\frac{1}{\omega_n r^{n-1}}.$$

For $n \geq 3$, integration gives

$$U(r) = \frac{1}{(n-2)\omega_n} r^{2-n} + C.$$

For $n = 2$, since $\omega_2 = 2\pi$, integration gives

$$U(r) = -\frac{1}{2\pi} \log r + C.$$

An additive constant is harmonic and does not affect the equation. Choosing it to be zero gives the normalized fundamental solution

$$\Gamma_n(x) = \begin{cases} -\frac{1}{2\pi} \log |x|, & n = 2, \\ \frac{1}{(n-2)\omega_n} |x|^{2-n}, & n \geq 3. \end{cases}$$

The sign is fixed by Gauss's law. The field $F = -\nabla \Gamma$ has outward flux 1, while $\nabla \Gamma$ has outward flux -1 . With the alternative convention $\Delta \Phi = \delta_0$, the fundamental solution is $\Phi = -\Gamma$.

4 Green functions for the Dirichlet problem

[Definition] For each fixed $x \in \Omega$, the Dirichlet Green function $G(x, y)$ satisfies, as a function of y ,

$$\begin{cases} -\Delta_y G(x, y) = \delta_x(y) & \text{in } \Omega, \\ G(x, y) = 0 & \text{for } y \in \partial\Omega. \end{cases}$$

The active variable in this definition is y : x is the pole and $-\Delta_y$ acts on $G(x, \cdot)$. By symmetry one also has $G(x, y) = G(y, x)$ under the usual assumptions, so an equivalent convention can act in x ; the notation must state which variable is differentiated.

Construction as singular part minus harmonic correction

Start from the whole-space fundamental solution $\Gamma(x - y)$. It has the correct singularity but does not vanish on the boundary. For fixed x , solve

$$\begin{cases} \Delta_y H_x(y) = 0 & \text{in } \Omega, \\ H_x(y) = \Gamma(x - y) & \text{on } \partial\Omega. \end{cases}$$

Then

$$G(x, y) = \Gamma(x - y) - H_x(y)$$

has both desired properties.

Green function of a ball

For the ball $B_R(0)$, Kelvin inversion gives the reflected point

$$x^* = \frac{R^2}{|x|^2} x.$$

For $n \geq 3$,

$$G_{B_R}(x, y) = \Gamma_n(x - y) - \left(\frac{R}{|x|}\right)^{n-2} \Gamma_n(x^* - y).$$

The two terms agree on $|y| = R$, so the boundary value is zero. The second pole lies outside the ball and hence contributes a harmonic correction inside.

5 Green's identities

[Lemma] The divergence theorem applied to $\phi \nabla \psi$ gives Green's first identity:

$$\int_{\Omega} (\nabla \phi \cdot \nabla \psi + \phi \Delta \psi) dx = \int_{\partial \Omega} \phi \partial_{\nu} \psi dS.$$

Interchanging ϕ and ψ and subtracting gives Green's second identity:

$$\int_{\Omega} (\phi \Delta \psi - \psi \Delta \phi) dx = \int_{\partial \Omega} (\phi \partial_{\nu} \psi - \psi \partial_{\nu} \phi) dS.$$

Taking $\phi = \psi = u$ in the first identity yields the energy relation

$$\int_{\Omega} |\nabla u|^2 dx = \int_{\partial \Omega} u \partial_{\nu} u dS - \int_{\Omega} u \Delta u dx.$$

For $u = 0$ on $\partial \Omega$,

$$\int_{\Omega} u(-\Delta u) dx = \int_{\Omega} |\nabla u|^2 dx,$$

which underlies uniqueness and positivity for the Dirichlet problem.

6 Green's representation formula

[Theorem] Assume

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial \Omega. \end{cases}$$

Then Green's representation formula is

$$u(x) = \int_{\Omega} G(x, y) f(y) dy - \int_{\partial \Omega} g(y) \frac{\partial G(x, y)}{\partial \nu_y} dS_y.$$

The boundary sign is negative for the convention $-\Delta G = \delta$. It comes from moving the term $-u(x)$ to the other side after applying Green's identity. Switching to the convention $\Delta G = \delta$ switches the related signs.

[Remark] Physical intuition. The representation formula separates the interior contribution from the boundary contribution.

The first term is the superposition of the contributions from all sources inside the domain. In electrostatic language, $f(y) dy$ is an infinitesimal charge at y , and $G(x, y)$ is the potential

measured at x from a unit charge at y while the boundary is grounded.

The second term is the contribution forced by the boundary. Prescribing $u = g$ on $\partial\Omega$ means prescribing the voltage of the boundary. In a conducting shell, electrons move until the required boundary voltage is attained. The resulting surface-charge distribution produces an additional harmonic potential inside Ω . The factor $-\partial_{\nu_y} G(x, y)$ measures how strongly the voltage imposed near the boundary point y influences the interior point x .

This normal derivative has a direct dipole interpretation. Fix $y \in \partial\Omega$ and let ν_y be the outward normal. Place a unit positive charge at

$$y_- = y - \frac{\varepsilon}{2}\nu_y$$

and a unit negative charge at

$$y_+ = y + \frac{\varepsilon}{2}\nu_y.$$

Using the whole-space fundamental solution $K(x, y) = \Gamma(x - y)$, the potential of this pair at x is

$$V_\varepsilon(x) = K(x, y_-) - K(x, y_+).$$

Taylor expansion in the source variable gives

$$K(x, y_\pm) = K(x, y) \pm \frac{\varepsilon}{2}\partial_{\nu_y} K(x, y) + O(\varepsilon^2).$$

Therefore

$$V_\varepsilon(x) = -\varepsilon\partial_{\nu_y} K(x, y) + O(\varepsilon^2).$$

The two unit charges have dipole moment of magnitude ε . Their potential tends to zero because their dipole moment tends to zero. Dividing by ε produces a dipole of unit moment:

$$\frac{V_\varepsilon(x)}{\varepsilon} \longrightarrow -\partial_{\nu_y} K(x, y).$$

The factor $1/\varepsilon$ has a charge-conservation interpretation. Suppose a fixed charge q per unit boundary area is distributed through a shell of thickness ε . If ρ_ε is approximately constant across the thickness, conservation of charge gives

$$q = \int_0^\varepsilon \rho_\varepsilon ds \simeq \varepsilon\rho_\varepsilon.$$

Therefore

$$\rho_\varepsilon \simeq \frac{q}{\varepsilon}.$$

As the shell becomes thinner, its density increases by exactly the reciprocal factor, so its integrated effect does not disappear. For a dipole layer, apply this concentration to two layers with opposite signs. Their total signed charge cancels, but their first moment survives. To prescribe dipole moment μ per unit area, the two surface-charge densities must satisfy

$$\sigma_\varepsilon^\pm = \pm \frac{\mu}{\varepsilon}, \quad \varepsilon|\sigma_\varepsilon^\pm| = |\mu|.$$

Thus division by the separation is the normalization that keeps the dipole moment finite. A pair of fixed unit charges would instead have moment ε and would vanish in the limit. The surviving boundary term is therefore the idealized charge-conserving shell limit together with fixed dipole-moment normalization.

For the Dirichlet Green function, use a point $y_\varepsilon = y - \varepsilon \nu_y$ lying inside the domain. Since $G(x, y) = 0$ on the boundary, a one-sided Taylor expansion gives

$$G(x, y_\varepsilon) = -\varepsilon \partial_{\nu_y} G(x, y) + O(\varepsilon^2).$$

Physically, the interior unit charge and the charge induced on the grounded boundary form a small dipole. After division by its separation ε , the limiting unit-dipole potential is

$$-\partial_{\nu_y} G(x, y).$$

Now distribute such normal dipoles over the boundary. If their dipole moment per unit area is $\mu(y)$, their total potential is

$$-\int_{\partial\Omega} \partial_{\nu_y} G(x, y) \mu(y) dS_y.$$

Comparison with Green's representation formula shows that the prescribed voltage $g(y)$ plays the role of the dipole-moment density:

$$\mu(y) = g(y).$$

With $\mu(y) = g(y)$, this calculation shows directly why the prescribed boundary voltage produces a finite nonzero contribution even when the idealized shell thickness tends to zero. This is an equivalent potential representation. On a physical conductor, the actual induced surface-charge density is instead determined by the normal electric field and is proportional, up to units and sign, to $-\partial_\nu u$.

Proof of the representation formula using Green's identity

[Proof] Fix $x \in \Omega$ and assume first that $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ and that $\partial\Omega$ is smooth. Because $G(x, y)$ is singular at $y = x$, Green's identity cannot be applied directly on all of Ω . Remove a small closed ball around the pole and set

$$\Omega_\varepsilon = \Omega \setminus \overline{B_\varepsilon(x)}.$$

On Ω_ε , both $u(y)$ and $G(x, y)$ are smooth and

$$\Delta_y G(x, y) = 0, \quad \Delta u(y) = -f(y).$$

Green's second identity, in the order needed for our sign convention, is

$$\int_{\Omega_\varepsilon} (u \Delta_y G - G \Delta u) dy = \int_{\partial\Omega_\varepsilon} (u \partial_\nu G - G \partial_\nu u) dS_y.$$

The left side is therefore

$$\int_{\Omega_\varepsilon} G(x, y) f(y) dy.$$

The boundary has two components:

$$\partial\Omega_\varepsilon = \partial\Omega \cup \partial B_\varepsilon(x).$$

On the outer boundary, the outward normal is the usual normal of Ω . Since $G = 0$ and $u = g$ on $\partial\Omega$,

$$\int_{\partial\Omega} (u\partial_\nu G - G\partial_\nu u) dS_y = \int_{\partial\Omega} g(y)\partial_{\nu_y} G(x, y) dS_y.$$

On the inner boundary, however, the outward unit normal of the punctured domain Ω_ε points into the removed ball:

$$\nu_{\Omega_\varepsilon} = -\frac{y-x}{|y-x|}.$$

Near the pole,

$$G(x, y) = \Gamma_n(y-x) + O(1),$$

where $-\Delta\Gamma_n = \delta_0$. Consequently,

$$\lim_{\varepsilon \downarrow 0} \int_{\partial B_\varepsilon(x)} \partial_{\nu_{\Omega_\varepsilon}} G(x, y) dS_y = 1.$$

Indeed, the flux with the radial normal pointing away from x is -1 , while the punctured-domain normal is the opposite normal. Since $u(y) = u(x) + O(\varepsilon)$,

$$\lim_{\varepsilon \downarrow 0} \int_{\partial B_\varepsilon(x)} u(y)\partial_{\nu_{\Omega_\varepsilon}} G(x, y) dS_y = u(x).$$

The other inner-boundary term vanishes:

$$\lim_{\varepsilon \downarrow 0} \int_{\partial B_\varepsilon(x)} G(x, y)\partial_{\nu_{\Omega_\varepsilon}} u(y) dS_y = 0.$$

For $n \geq 3$, this is because $G = O(\varepsilon^{2-n})$, $\partial_\nu u = O(1)$, and $|\partial B_\varepsilon| = O(\varepsilon^{n-1})$, so the integral is $O(\varepsilon)$. For $n = 2$, it is $O(\varepsilon|\log \varepsilon|)$.

Taking $\varepsilon \downarrow 0$ in Green's identity now gives

$$\int_{\Omega} G(x, y)f(y) dy = \int_{\partial\Omega} g(y)\partial_{\nu_y} G(x, y) dS_y + u(x).$$

Solving for $u(x)$ proves

$$u(x) = \int_{\Omega} G(x, y)f(y) dy - \int_{\partial\Omega} g(y)\partial_{\nu_y} G(x, y) dS_y.$$

The punctured-domain proof makes the inner-boundary orientation and the sign explicit.

Day 2

7 The Dirichlet problem on a ball

[Notation] Let

$$B_R(0) = \{x \in \mathbb{R}^n : |x| < R\}$$

be the n -dimensional ball. Physically, its boundary $\partial B_R(0)$ is a conducting shell. Prescribing the voltage on the shell means prescribing a boundary function g . If the interior charge density is zero, then the potential is harmonic, so the problem is

$$\begin{cases} \Delta u = 0 & \text{in } B_R(0), \\ u = g & \text{on } \partial B_R(0). \end{cases}$$

[Theorem] For an interior point $x \in B_R(0)$, Green's representation formula gives

$$u(x) = \int_{\partial B_R(0)} \frac{R^2 - |x|^2}{\omega_n R |x - y|^n} g(y) dS_y,$$

where $\omega_n = |S^{n-1}|$. Thus the potential at x is a weighted average of the voltages applied to the shell. Boundary points nearer to x receive a larger weight through the factor $|x - y|^{-n}$. The weight is normalized because

$$\int_{\partial B_R(0)} \frac{R^2 - |x|^2}{\omega_n R |x - y|^n} dS_y = 1.$$

At the center $x = 0$, every point on the shell is the same distance R from the center. The weight therefore becomes constant:

$$\frac{R^2}{\omega_n R |y|^n} = \frac{1}{\omega_n R^{n-1}} = \frac{1}{|\partial B_R(0)|}.$$

Consequently,

$$u(0) = \frac{1}{|\partial B_R(0)|} \int_{\partial B_R(0)} g(y) dS_y.$$

Since $u = g$ on the boundary, this may also be written as

$$u(0) = \frac{1}{|\partial B_R(0)|} \int_{\partial B_R(0)} u(y) dS_y.$$

[Corollary] This is the mean-value property at the center: the value of a harmonic function at the center of a ball equals its average over the boundary sphere. After translating the ball, the general form is

$$u(x_0) = \frac{1}{|\partial B_r(x_0)|} \int_{\partial B_r(x_0)} u(y) dS_y$$

whenever $B_r(x_0) \Subset \Omega$.

8 Characterizations by the mean-value property

[Theorem] Let $\Omega \subset \mathbb{R}^n$ be open and let $u \in C(\Omega)$. The following statements are equivalent.

1. The function u belongs to $C^2(\Omega)$ and is harmonic:

$$\Delta u = 0 \quad \text{in } \Omega.$$

2. The function satisfies the spherical mean-value property: for every $B_r(x) \Subset \Omega$,

$$u(x) = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u(y) dS_y.$$

3. The function satisfies the ball mean-value property: for every $B_r(x) \Subset \Omega$,

$$u(x) = \frac{1}{|B_r(x)|} \int_{B_r(x)} u(y) dy.$$

Harmonicity implies the spherical mean-value property

[Proof] Fix $x \in \Omega$ and define the spherical average

$$A(r) = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u(y) dS_y = \frac{1}{\omega_n} \int_{S^{n-1}} u(x + r\theta) dS_\theta.$$

Differentiating with respect to r gives

$$\begin{aligned} A'(r) &= \frac{1}{\omega_n} \int_{S^{n-1}} \nabla u(x + r\theta) \cdot \theta dS_\theta \\ &= \frac{1}{\omega_n r^{n-1}} \int_{\partial B_r(x)} \partial_\nu u dS \\ &= \frac{1}{\omega_n r^{n-1}} \int_{B_r(x)} \Delta u dy \\ &= 0. \end{aligned}$$

Thus $A(r)$ is constant, not merely monotone. By continuity,

$$\lim_{r \downarrow 0} A(r) = u(x),$$

so $A(r) = u(x)$. This proves the spherical mean-value property. Equivalently, one can apply the ball representation formula with the evaluation point at the center.

The spherical property implies the ball property

Using polar coordinates and the spherical mean-value property,

$$\begin{aligned}
\int_{B_R(x)} u(y) dy &= \int_0^R \left(\int_{\partial B_r(x)} u(y) dS_y \right) dr \\
&= \int_0^R |\partial B_r(x)| u(x) dr \\
&= |B_R(x)| u(x).
\end{aligned}$$

Dividing by $|B_R(x)|$ proves the ball mean-value property.

The ball property implies harmonicity

Let $\rho \in C_c^\infty(\mathbb{R}^n)$ be a nonnegative radial mollifier satisfying

$$\int_{\mathbb{R}^n} \rho(y) dy = 1,$$

and set

$$\rho_\varepsilon(y) = \varepsilon^{-n} \rho\left(\frac{y}{\varepsilon}\right).$$

For x whose distance from $\partial\Omega$ is greater than ε , radial integration and the ball mean-value property give

$$\begin{aligned}
(u * \rho_\varepsilon)(x) &= \int_{\mathbb{R}^n} u(x-y) \rho_\varepsilon(y) dy \\
&= u(x) \int_{\mathbb{R}^n} \rho_\varepsilon(y) dy \\
&= u(x).
\end{aligned}$$

The second equality follows because a radial mollifier is a weighted superposition of averages over balls centered at x . Since $u * \rho_\varepsilon$ is smooth, the equality $u = u * \rho_\varepsilon$ locally shows that

$$u \in C^\infty(\Omega).$$

It remains to show that $\Delta u = 0$. Fix $x \in \Omega$ and set

$$F(r) = \int_{B_r(x)} u(y) dy.$$

The ball mean-value property gives

$$F(r) = |B_r| u(x).$$

Differentiating with respect to r yields

$$\int_{\partial B_r(x)} u(y) dS_y = |\partial B_r| u(x).$$

Thus the spherical average

$$A(r) = \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} u(y) dS_y$$

is equal to $u(x)$ and hence is constant in r . Differentiating the spherical average and applying the divergence theorem gives

$$0 = A'(r) = \frac{1}{\omega_n r^{n-1}} \int_{B_r(x)} \Delta u(y) dy.$$

Therefore

$$\frac{1}{|B_r|} \int_{B_r(x)} \Delta u(y) dy = 0.$$

Because Δu is continuous, letting $r \downarrow 0$ gives

$$\Delta u(x) = 0.$$

Since x was arbitrary, u is harmonic in Ω .

9 Maximum principles

Strong maximum principle

[Theorem] Let $\Omega \subset \mathbb{R}^n$ be connected, and let u be harmonic in Ω . If u attains its maximum at an interior point, then u is constant on Ω .

[Proof] Suppose

$$u(x_0) = M = \sup_{\Omega} u$$

for some $x_0 \in \Omega$. Define the maximum set

$$E = \{x \in \Omega : u(x) = M\}.$$

The set E is nonempty because $x_0 \in E$, and it is closed relative to Ω because u is continuous.

To show that E is open, take $x \in E$ and choose $r > 0$ such that

$$B_r(x) \Subset \Omega.$$

The ball mean-value property gives

$$M = u(x) = \frac{1}{|B_r(x)|} \int_{B_r(x)} u(y) dy.$$

Since $u(y) \leq M$ throughout the ball, its average can equal M only if

$$u(y) = M$$

for every $y \in B_r(x)$. Thus $B_r(x) \subset E$, so E is open.

Therefore E is nonempty and both open and closed in the connected set Ω . Hence

$$E = \Omega,$$

which means that u is constant.

Applying the theorem to $-u$ gives the corresponding strong minimum principle. Consequently, if $|u|$ attains its maximum at an interior point, then u is constant as well.

Weak maximum principle

[Theorem] Let $\Omega \subset \mathbb{R}^n$ be bounded and let

$$u \in C^2(\Omega) \cap C(\overline{\Omega}).$$

If $\Delta u \geq 0$ in Ω , then

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u.$$

If $\Delta u \leq 0$ in Ω , then

$$\min_{\overline{\Omega}} u = \min_{\partial\Omega} u.$$

In particular, both conclusions hold when u is harmonic.

The domain need not be connected. Boundedness and continuity on $\overline{\Omega}$ ensure that the extrema are attained.

[Proof] Assume first that $\Delta u \geq 0$. Since Ω is bounded, choose R such that $\Omega \subset B_R(0)$. For $\varepsilon > 0$, set

$$u_\varepsilon(x) = u(x) + \varepsilon|x|^2.$$

Then $\Delta u_\varepsilon > 0$, so u_ε cannot attain its maximum in the interior. Consequently, for every $x \in \Omega$,

$$u(x) \leq u_\varepsilon(x) \leq \max_{\partial\Omega} u + \varepsilon R^2.$$

Letting $\varepsilon \downarrow 0$ proves the maximum statement. Applying it to $-u$ proves the minimum statement when $\Delta u \leq 0$.

Uniqueness of the Dirichlet problem

[Corollary] The Dirichlet problem has at most one solution in $C^2(\Omega) \cap C(\overline{\Omega})$.

Indeed, if u_1 and u_2 have the same Laplacian and the same boundary values, then

$$w = u_1 - u_2$$

satisfies

$$\Delta w = 0 \quad \text{in } \Omega, \quad w = 0 \quad \text{on } \partial\Omega.$$

The weak maximum principle applied to w and $-w$ gives

$$w \leq 0, \quad -w \leq 0.$$

Thus $w = 0$, and therefore $u_1 = u_2$.

10 Interior gradient estimates and Liouville's theorem

Gradient estimate on a ball

[Theorem] Suppose that u is harmonic in $B_R(x_0)$. Then

$$|\nabla u(x_0)| \leq \frac{n}{R} \sup_{B_R(x_0)} |u|.$$

[Proof] Fix $0 < r < R$. For x sufficiently close to x_0 , the ball mean-value property gives

$$u(x) = \frac{1}{|B_r|} \int_{B_r(0)} u(x+z) dz.$$

Differentiate this identity with respect to the center x . Differentiation under the integral sign and the divergence theorem give

$$\begin{aligned} \nabla u(x) &= \frac{1}{|B_r|} \int_{B_r(0)} \nabla u(x+z) dz \\ &= \frac{1}{|B_r|} \int_{\partial B_r(0)} u(x+z) \nu(z) dS_z. \end{aligned}$$

Evaluating at $x = x_0$ and taking norms gives

$$\begin{aligned} |\nabla u(x_0)| &\leq \frac{|\partial B_r|}{|B_r|} \sup_{B_r(x_0)} |u| \\ &= \frac{n}{r} \sup_{B_r(x_0)} |u|. \end{aligned}$$

Letting $r \uparrow R$ gives the stated estimate.

Liouville's theorem

[Corollary] Every bounded harmonic function on $\mathbb{R}^n$ is constant.

[Proof] Suppose

$$|u(x)| \leq M$$

for all $x \in \mathbb{R}^n$. Fix $x_0 \in \mathbb{R}^n$. Since u is harmonic on $B_R(x_0)$ for every $R > 0$, the interior gradient estimate gives

$$|\nabla u(x_0)| \leq \frac{nM}{R}.$$

Letting $R \rightarrow \infty$ yields

$$\nabla u(x_0) = 0.$$

Because x_0 was arbitrary, $\nabla u = 0$ on $\mathbb{R}^n$, and hence u is constant.

11 Compactness of families of harmonic functions

[Theorem] Let $\Omega \subset \mathbb{R}^n$ be open, and let $\{u_k\}$ be a sequence of harmonic functions on Ω . Suppose the sequence is locally uniformly bounded: for every compact set $K \subset \Omega$, there is a constant C_K such that

$$\sup_k \sup_K |u_k| \leq C_K.$$

Then there are a subsequence $\{u_{k_j}\}$ and a harmonic function u on Ω such that

$$u_{k_j} \longrightarrow u$$

locally uniformly on Ω .

[Proof] Choose a compact exhaustion

$$K_1 \Subset K_2 \Subset K_3 \Subset \cdots, \quad \bigcup_{j=1}^{\infty} K_j = \Omega.$$

For each j , define

$$d_j = \text{dist}(K_j, \Omega \setminus \text{int } K_{j+1}) > 0.$$

Local uniform boundedness provides a constant C_{j+1} such that

$$\sup_k \sup_{K_{j+1}} |u_k| \leq C_{j+1}.$$

The interior gradient estimate then gives

$$\sup_k \sup_{K_j} |\nabla u_k| \leq \frac{nC_{j+1}}{d_j}.$$

Thus $\{u_k\}$ is uniformly bounded and equicontinuous on each K_j . By the Arzelà–Ascoli theorem, there is a subsequence converging uniformly on K_1 . From this subsequence, choose a further subsequence converging uniformly on K_2 , and continue inductively.

Taking the diagonal subsequence produces functions u_{k_j} and a continuous function u such that

$$u_{k_j} \longrightarrow u$$

uniformly on every compact subset of Ω .

It remains to check that the limit is harmonic. Fix a ball

$$B_r(x) \Subset \Omega.$$

Every u_{k_j} satisfies the spherical mean-value property:

$$u_{k_j}(x) = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u_{k_j}(y) dS_y.$$

The convergence is uniform on the compact sphere $\partial B_r(x)$, so we may pass to the limit and obtain

$$u(x) = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u(y) dS_y.$$

Therefore u satisfies the spherical mean-value property on every $B_r(x) \Subset \Omega$. By the mean-value characterization, u is harmonic.

12 Subharmonic functions

[Definition] A function $v \in C(\Omega)$ is called subharmonic if it satisfies the following harmonic-comparison property. For every ball B with

$$B \Subset \Omega$$

and every function

$$h \in C(\overline{B}) \cap C^2(B), \quad \Delta h = 0 \quad \text{in } B,$$

the boundary inequality

$$v \leq h \quad \text{on } \partial B$$

must imply

$$v \leq h \quad \text{throughout } B.$$

Thus a subharmonic function cannot cross from below a harmonic function while the harmonic function dominates it on the boundary. A function v is superharmonic when $-v$ is subharmonic.

[Theorem] If $v \in C^2(\Omega)$, then

$$v \text{ is subharmonic} \iff \Delta v \geq 0 \text{ in } \Omega.$$

Nonnegative Laplacian implies harmonic comparison

[Proof] Assume $\Delta v \geq 0$. Let $B = B_R(a) \Subset \Omega$, and let h be harmonic in B , continuous on $\overline{B}$, and satisfy

$$v \leq h \quad \text{on } \partial B.$$

Set $w = v - h$. Then

$$\Delta w = \Delta v - \Delta h \geq 0, \quad w \leq 0 \quad \text{on } \partial B.$$

By the weak maximum principle,

$$\max_{\overline{B}} w = \max_{\partial B} w \leq 0.$$

Hence $w \leq 0$ in B , so $v \leq h$ throughout B . Therefore v is subharmonic.

Harmonic comparison implies nonnegative Laplacian

Conversely, suppose that v is subharmonic in the comparison sense. Fix a ball

$$B_r(x) \Subset \Omega,$$

and let h be the harmonic function on $B_r(x)$ whose boundary values agree with v :

$$\Delta h = 0 \quad \text{in } B_r(x), \quad h = v \quad \text{on } \partial B_r(x).$$

Such an h is given by the ball representation formula. The comparison property gives

$$v \leq h \quad \text{in } B_r(x).$$

At the center, the mean-value property for h yields

$$v(x) \leq h(x) = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} h(y) dS_y = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} v(y) dS_y.$$

Thus the harmonic-comparison definition implies the spherical mean-value inequality. Taylor expansion at x and symmetry of the sphere give

$$\frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} v(y) dS_y = v(x) + \frac{r^2}{2n} \Delta v(x) + o(r^2).$$

The subharmonic mean-value inequality therefore implies

$$0 \leq \frac{r^2}{2n} \Delta v(x) + o(r^2).$$

Divide by r^2 and let $r \downarrow 0$. Then

$$\Delta v(x) \geq 0.$$

Since x was arbitrary, $\Delta v \geq 0$ throughout Ω .

Day 3

[Remark] For a general domain, finding its Green function explicitly is very difficult. For a ball, however, the Green function and the corresponding Dirichlet solution are known. The Perron method exploits these local solutions on balls to construct a solution on the full domain without first finding a global Green function.

13 Perron method

The Perron method constructs harmonic functions from suitable families of subharmonic functions. We first record four properties that make this construction possible.

[Lemma] Let v and w be subharmonic on Ω .

1. The functions $v + w$ and av , for every constant $a \geq 0$, are subharmonic on Ω .
2. The function $\max\{v, w\}$ is subharmonic on Ω .
3. Suppose Ω is bounded and v satisfies

$$\limsup_{\substack{x \rightarrow \zeta \\ x \in \Omega}} v(x) \leq 0 \quad \text{for every } \zeta \in \partial\Omega.$$

Then $v \leq 0$ in Ω .

4. Let $D \subset \Omega$ be open, and let g be subharmonic on D . Suppose

$$\limsup_{\substack{x \rightarrow \zeta \\ x \in D}} g(x) \leq v(\zeta) \quad \text{for every } \zeta \in \partial D \cap \Omega.$$

Define

$$\tilde{v}(x) = \begin{cases} \max\{v(x), g(x)\}, & x \in D, \\ v(x), & x \in \Omega \setminus D. \end{cases}$$

Then $\tilde{v}$ is subharmonic on Ω .

[Proof] We prove the four statements in order.

For 1, let $B \Subset \Omega$ be a ball and let h be harmonic on B , continuous on $\overline{B}$, with

$$v + w \leq h \quad \text{on } \partial B.$$

Let h_v be the harmonic function on B with boundary values v . Since v is subharmonic,

$$v \leq h_v \quad \text{in } B.$$

On ∂B we have $h_v = v$, and hence

$$w \leq h - v = h - h_v.$$

The function $h - h_v$ is harmonic, so the comparison property for w gives

$$w \leq h - h_v \quad \text{in } B.$$

Adding the two inequalities yields $v + w \leq h$ in B . Thus $v + w$ is subharmonic. If $a > 0$ and $av \leq h$ on ∂B , then $v \leq h/a$ there. Since h/a is harmonic, $v \leq h/a$ in B , so $av \leq h$. The case $a = 0$ is immediate.

For 2, suppose

$$\max\{v, w\} \leq h \quad \text{on } \partial B,$$

where h is harmonic on B . Then both $v \leq h$ and $w \leq h$ on ∂B . Comparison gives both inequalities throughout B , and therefore

$$\max\{v, w\} \leq h \quad \text{in } B.$$

For 3, first recall that the comparison definition implies the spherical mean-value inequality. Indeed, if $B_r(x) \Subset \Omega$ and h is the harmonic function with boundary values v , then

$$v(x) \leq h(x) = \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} v(y) dS_y.$$

Suppose for contradiction that

$$M := \sup_{\Omega} v > 0.$$

Since Ω is bounded and the boundary limsup is nonpositive, there is a compact set $K \Subset \Omega$ such that

$$v < \frac{M}{2} \quad \text{on } \Omega \setminus K.$$

Consequently, v attains its maximum M at some $x_0 \in K$. Let Ω_0 be the connected component of Ω containing x_0 , and set

$$E = \{x \in \Omega_0 : v(x) = M\}.$$

The set E is closed in Ω_0 . It is also open. To see this, take $x \in E$ and choose $r > 0$ such that $B_r(x) \Subset \Omega_0$. For every $0 < \rho < r$, the mean-value inequality gives

$$M = v(x) \leq \frac{1}{|\partial B_\rho(x)|} \int_{\partial B_\rho(x)} v(y) dS_y \leq M.$$

Thus equality holds. Since $v \leq M$ and v is continuous, $v = M$ on every sphere $\partial B_\rho(x)$, and hence on $B_r(x)$. Therefore E is open. Connectedness gives $E = \Omega_0$. Since Ω_0 is bounded, there is a sequence in Ω_0 converging to some point of $\partial\Omega_0 \subset \partial\Omega$, which contradicts the boundary limsup condition. Hence $M \leq 0$, and therefore $v \leq 0$ in Ω .

For 4, the boundary limsup assumption and the continuity of v show that $\tilde{v}$ is continuous across $\partial D \cap \Omega$. It remains to verify harmonic comparison. Let $B \Subset \Omega$ be a ball, and let h be harmonic on B , continuous on $\overline{B}$, with

$$\tilde{v} \leq h \quad \text{on } \partial B.$$

Since $v \leq \tilde{v}$ on ∂B and v is subharmonic, comparison gives

$$v \leq h \quad \text{in } B.$$

We next prove $g \leq h$ on $D \cap B$. The function $g - h$ is subharmonic there. Its boundary has two relevant pieces.

On $D \cap \partial B$,

$$g \leq \tilde{v} \leq h.$$

On $B \cap \partial D$, the gluing assumption and the inequality already proved for v give

$$\limsup_{\substack{x \rightarrow \zeta \\ x \in D \cap B}} (g(x) - h(x)) \leq v(\zeta) - h(\zeta) \leq 0.$$

At a point of $\partial D \cap \partial B$, the same conclusion follows by combining these two limiting inequalities. Thus

$$\limsup_{\substack{x \rightarrow \zeta \\ x \in D \cap B}} (g(x) - h(x)) \leq 0 \quad \text{for every } \zeta \in \partial(D \cap B).$$

Property 3, applied to the bounded open set $D \cap B$, yields

$$g - h \leq 0 \quad \text{in } D \cap B.$$

Hence $g \leq h$ on $D \cap B$. Together with $v \leq h$ in B , this gives

$$\tilde{v} \leq h \quad \text{throughout } B.$$

Therefore $\tilde{v}$ is subharmonic on Ω .

14 Perron class, Perron function, and harmonic lifting

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain, and let

$$\varphi \in C(\partial\Omega)$$

be the prescribed boundary data.

[Definition] The Perron class associated with φ is

$$\mathcal{S}_\varphi = \left\{ v \in C(\Omega) : \begin{array}{l} v \text{ is subharmonic on } \Omega, \\ v \text{ is bounded from above, and} \\ \limsup_{\substack{x \rightarrow \zeta \\ x \in \Omega}} v(x) \leq \varphi(\zeta) \text{ for every } \zeta \in \partial\Omega \end{array} \right\}.$$

Its elements are the subharmonic lower competitors for the Dirichlet problem with boundary data φ .

[Definition] The Perron function is the pointwise supremum

$$U_\varphi(x) = \sup_{v \in \mathcal{S}_\varphi} v(x), \quad x \in \Omega.$$

[Lemma] The Perron class is nonempty, and U_φ is finite everywhere. More precisely, if

$$m = \min_{\partial\Omega} \varphi, \quad M = \max_{\partial\Omega} \varphi,$$

then

$$m \leq U_\varphi(x) \leq M \quad \text{for every } x \in \Omega.$$

[Proof] The constant function m is harmonic, hence subharmonic, and satisfies the required boundary inequality. Thus

$$m \in \mathcal{S}_\varphi,$$

so the class is nonempty and $U_\varphi \geq m$.

Now take $v \in \mathcal{S}_\varphi$. The function $v - M$ is subharmonic and satisfies

$$\limsup_{\substack{x \rightarrow \zeta \\ x \in \Omega}} (v(x) - M) \leq \varphi(\zeta) - M \leq 0$$

for every $\zeta \in \partial\Omega$. Property 3 of the preceding lemma gives

$$v \leq M \quad \text{in } \Omega.$$

Taking the supremum over $v \in \mathcal{S}_\varphi$ yields $U_\varphi \leq M$.

[Definition] Let v be subharmonic on Ω , let $B = B_r(x_0) \Subset \Omega$, and denote by H_v^B the solution of the Dirichlet problem on B with boundary values v . The harmonic lifting of v over B is defined by

$$\Delta H_v^B = 0 \quad \text{in } B, \quad H_v^B = v \quad \text{on } \partial B,$$

$$v^B(x) = \begin{cases} H_v^B(x), & x \in B, \\ v(x), & x \in \Omega \setminus B. \end{cases}$$

Thus harmonic lifting replaces v inside the ball by the harmonic function with the same boundary values and leaves v unchanged outside the ball.

[Lemma] Let v be subharmonic on Ω and $B \Subset \Omega$ be a ball. Then:

1. $v^B \in C(\Omega)$.
2. v^B is subharmonic on Ω .
3. $v^B \geq v$ on Ω .
4. If $v \in \mathcal{S}_\varphi$, then $v^B \in \mathcal{S}_\varphi$.

[Proof] Since $H_v^B = v$ on ∂B , the two pieces in the definition of v^B agree continuously along ∂B . This proves 1.

Because v is subharmonic and H_v^B is harmonic with the same boundary values, harmonic comparison gives

$$v \leq H_v^B \quad \text{in } B.$$

Therefore $v^B \geq v$, proving 3. Moreover, on B we have

$$v^B = H_v^B = \max\{v, H_v^B\}.$$

The pasting property from the preceding lemma, applied with $D = B$ and $g = H_v^B$, now shows that v^B is subharmonic on Ω . This proves 2.

Finally, suppose $v \in \mathcal{S}_\varphi$. Since $\bar{B} \Subset \Omega$, the lifting does not change v near $\partial\Omega$. Consequently,

$$\limsup_{\substack{x \rightarrow \zeta \\ x \in \Omega}} v^B(x) = \limsup_{\substack{x \rightarrow \zeta \\ x \in \Omega}} v(x) \leq \varphi(\zeta)$$

for every $\zeta \in \partial\Omega$. The function v^B remains bounded from above because both v and H_v^B are bounded from above. Together with 1 and 2, this proves $v^B \in \mathcal{S}_\varphi$.

[Theorem] The Perron function U_φ is harmonic on Ω .

[Proof] Fix $x_0 \in \Omega$ and a ball

$$B = B_r(x_0) \Subset \Omega.$$

We prove that U_φ is harmonic on B .

Step 1. Construct a harmonic candidate. By the definition of the supremum, choose $v_j \in \mathcal{S}_\varphi$ such that

$$U_\varphi(x_0) - \frac{1}{j} < v_j(x_0) \leq U_\varphi(x_0).$$

Set

$$\tilde{v}_j = \max\{v_j, m\},$$

where $m = \min_{\partial\Omega} \varphi$. The maximum property for subharmonic functions shows that $\tilde{v}_j \in \mathcal{S}_\varphi$, and

$$m \leq \tilde{v}_j \leq U_\varphi \leq M, \quad \tilde{v}_j(x_0) \longrightarrow U_\varphi(x_0).$$

Harmonically lift $\tilde{v}_j$ over B and write

$$h_j = (\tilde{v}_j)^B.$$

Each h_j belongs to $\mathcal{S}_\varphi$, is harmonic on B , and satisfies

$$m \leq h_j \leq U_\varphi \leq M.$$

The compactness theorem for locally uniformly bounded harmonic functions gives a subsequence, still denoted by h_j , converging locally uniformly on B to a harmonic function h . Since harmonic lifting increases a subharmonic function,

$$\tilde{v}_j(x_0) \leq h_j(x_0) \leq U_\varphi(x_0).$$

Therefore

$$h(x_0) = U_\varphi(x_0).$$

Moreover, $h_j \leq U_\varphi$ because $h_j \in \mathcal{S}_\varphi$. Passing to the limit gives

$$h \leq U_\varphi \quad \text{on } B.$$

Step 2. Prove the reverse inequality. Suppose for contradiction that there is a point $z \in B$ such that

$$h(z) < U_\varphi(z).$$

By the definition of $U_\varphi(z)$, there exists $f \in \mathcal{S}_\varphi$ such that

$$h(z) < f(z) \leq U_\varphi(z).$$

Define

$$g_j = \max\{f, \tilde{v}_j\} \in \mathcal{S}_\varphi,$$

and harmonically lift once more:

$$k_j = (g_j)^B.$$

Then $k_j \in \mathcal{S}_\varphi$, k_j is harmonic on B , and

$$m \leq k_j \leq M.$$

After passing to a further subsequence, the harmonic compactness theorem gives

$$k_j \longrightarrow k$$

locally uniformly on B , where k is harmonic.

On ∂B we have $g_j \geq \tilde{v}_j$. Comparing their harmonic extensions gives

$$k_j \geq h_j \quad \text{in } B.$$

Since $k_j \in \mathcal{S}_\varphi$, we also have $k_j \leq U_\varphi$. At x_0 this yields

$$U_\varphi(x_0) \geq k_j(x_0) \geq h_j(x_0).$$

Taking limits gives

$$k(x_0) = h(x_0) = U_\varphi(x_0),$$

while $k \geq h$ throughout B . Thus $k - h$ is a nonnegative harmonic function that vanishes at the interior point x_0 . By the strong minimum principle,

$$k = h \quad \text{on } B.$$

On the other hand, harmonic lifting dominates the function being lifted, so

$$k_j \geq g_j \geq f \quad \text{in } B.$$

Passing to the limit at z gives

$$k(z) \geq f(z) > h(z),$$

contradicting $k = h$. Hence $U_\varphi \leq h$ on B . Together with the opposite inequality, this gives

$$U_\varphi = h \quad \text{on } B.$$

Therefore U_φ is harmonic in a neighbourhood of the arbitrary point x_0 , and hence harmonic on all of Ω .

15 Barriers and boundary values

The preceding theorem proves harmonicity in the interior. To solve the Dirichlet problem, we must still show that the Perron function approaches the prescribed value at each boundary point.

[Definition] Let $\zeta \in \partial\Omega$. A barrier at ζ is a function

$$w_\zeta \in C(\overline{\Omega}) \cap C^2(\Omega)$$

such that

$$\Delta w_\zeta \leq 0 \quad \text{in } \Omega, \quad w_\zeta(\zeta) = 0, \quad w_\zeta(x) > 0 \quad \text{for } x \in \overline{\Omega} \setminus \{\zeta\}.$$

Thus a barrier is superharmonic, touches zero only at the chosen boundary point, and is strictly positive everywhere else.

[Theorem] Suppose every point $\zeta \in \partial\Omega$ admits a barrier. Then, for every $\varphi \in C(\partial\Omega)$, the Perron function satisfies

$$\lim_{\substack{x \rightarrow \zeta \\ x \in \Omega}} U_\varphi(x) = \varphi(\zeta) \quad \text{for every } \zeta \in \partial\Omega.$$

Consequently, U_φ extends continuously to $\overline{\Omega}$ with boundary values φ and solves the Dirichlet problem.

[Proof] Fix $\zeta \in \partial\Omega$ and a barrier w_ζ . Let $\varepsilon > 0$, and set

$$M = \max_{\partial\Omega} |\varphi|.$$

By continuity of φ , there is a relatively open neighbourhood N_ε of ζ in $\partial\Omega$ such that

$$|\varphi(\eta) - \varphi(\zeta)| < \varepsilon \quad \text{for every } \eta \in N_\varepsilon.$$

The set

$$F_\varepsilon = \partial\Omega \setminus N_\varepsilon$$

is compact and does not contain ζ . Since the barrier is strictly positive away from ζ ,

$$c_\varepsilon := \min_{F_\varepsilon} w_\zeta > 0.$$

We first construct an upper comparison function:

$$\overline{u}_\varepsilon(x) = \varphi(\zeta) + \varepsilon + A w_\zeta(x).$$

This function is superharmonic. Let $v \in \mathcal{S}_\varphi$. For $\eta \in N_\varepsilon$,

$$\begin{aligned} \limsup_{\substack{x \rightarrow \eta \\ x \in \Omega}} (v(x) - \overline{u}_\varepsilon(x)) &\leq \varphi(\eta) - \varphi(\zeta) - \varepsilon - A w_\zeta(\eta) \\ &\leq 0. \end{aligned}$$

Now choose $A > 0$ sufficiently large, depending on ε , so that

$$Ac_\varepsilon \geq 2M.$$

For $\eta \in F_\varepsilon$, we then have

$$\begin{aligned} \limsup_{\substack{x \rightarrow \eta \\ x \in \Omega}} (v(x) - \bar{u}_\varepsilon(x)) &\leq 2M - \varepsilon - Ac_\varepsilon \\ &\leq 0. \end{aligned}$$

Since $v - \bar{u}_\varepsilon$ is subharmonic, property 3 gives

$$v \leq \bar{u}_\varepsilon \quad \text{in } \Omega.$$

Taking the supremum over the Perron class yields

$$U_\varphi(x) \leq \varphi(\zeta) + \varepsilon + Aw_\zeta(x).$$

Because $w_\zeta(x) \rightarrow 0$ as $x \rightarrow \zeta$,

$$\limsup_{\substack{x \rightarrow \zeta \\ x \in \Omega}} U_\varphi(x) \leq \varphi(\zeta) + \varepsilon.$$

For the lower estimate, define

$$\underline{u}_\varepsilon(x) = \varphi(\zeta) - \varepsilon - Aw_\zeta(x).$$

This function is subharmonic. On N_ε ,

$$\underline{u}_\varepsilon \leq \varphi(\zeta) - \varepsilon \leq \varphi,$$

while on F_ε ,

$$\underline{u}_\varepsilon \leq M - \varepsilon - Ac_\varepsilon \leq -M \leq \varphi.$$

Thus $\underline{u}_\varepsilon \in \mathcal{S}_\varphi$, so

$$\varphi(\zeta) - \varepsilon - Aw_\zeta(x) \leq U_\varphi(x).$$

Letting $x \rightarrow \zeta$ gives

$$\varphi(\zeta) - \varepsilon \leq \liminf_{\substack{x \rightarrow \zeta \\ x \in \Omega}} U_\varphi(x).$$

Combining the upper and lower estimates and then letting $\varepsilon \downarrow 0$ proves

$$\lim_{\substack{x \rightarrow \zeta \\ x \in \Omega}} U_\varphi(x) = \varphi(\zeta).$$

[Lemma] Suppose Ω satisfies the exterior sphere condition at $\zeta \in \partial\Omega$: there are $y \notin \overline{\Omega}$ and $R > 0$ such that

$$B_R(y) \cap \Omega = \emptyset, \quad \overline{B_R(y)} \cap \overline{\Omega} = \{\zeta\}.$$

Then ζ admits a barrier.

[Proof] Since every $x \in \overline{\Omega} \setminus \{\zeta\}$ satisfies $|x - y| > R$, define

$$w_\zeta(x) = \begin{cases} R^{2-n} - |x - y|^{2-n}, & n \geq 3, \\ \log \frac{|x - y|}{R}, & n = 2. \end{cases}$$

The function is harmonic on Ω , continuous on $\overline{\Omega}$, vanishes at ζ , and is strictly positive at every other point of $\overline{\Omega}$. Hence it is a barrier.

[Corollary] Every bounded domain with C^2 boundary admits a barrier at each boundary point. Therefore, for every continuous boundary datum φ , its Perron function belongs to

$$C(\overline{\Omega}) \cap C^2(\Omega),$$

is harmonic in Ω , and equals φ on $\partial\Omega$.

[Proof] A C^2 boundary satisfies the exterior sphere condition at every boundary point, so the preceding lemma and theorem apply.